Fiscal Year:	FY 2020	**Task Last Updated:**	FY 08/03/2020
PI Name:	Lai, Matteo M.Sc.		
Project Title:	EmbraceX		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	TRISH--TRISH		
Joint Agency Name:		**TechPort:**	Yes
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	02108	**Congressional District:**	8
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	TRISH--Industry
Start Date:	05/01/2019	**End Date:**	04/30/2020
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	TRISH
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNX16AO69A-IND0010		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	Industry Project This project proposes to develop a wrist-worn wearable device that monitors, among other things, electrodermal activity (EDA), accelerometry, heart rate, SpO2 (oxygen saturation), and peripheral temperature sensing. The development process will follow a traditional wearable consumer electronics development process where they will prove out the functionality of subsystems (Proof-Of-Concept), integrate into a Form-factor system (Alpha), and bring to mass production via Engineering Validation (EV) Design Validation (DV) and Process Validation (PV) builds.		
Rationale for HRP Directed Research:			

Research Impact/Earth Benefits:	We can distinguish between internal spillover effects that have been generated during the engineering development process and benefits for the broad research community enabled by the final product once it reaches commercialization. From an internal Empatica perspective, EmbracePlus is an example of combinatorial innovation, where innovation happens at the crossroads of market needs and the capability of a company to integrate and combine different modules/system subparts, in an adequate form factor that meets the requirements of the end user. The Translational Research Institute for Space Health (TRISH)/NASA contribution has allowed Empatica to deepen the level of systemic and technological integration, allowing the development teams to explore more combinations and test different options in order to achieve the best balance between usability, performance, and technical/design constraints. Moreover, the EmbracePlus engineering development process has been enhanced by the challenge of considering usability in the context of space missions. This constraint influenced the design of the product, the choice of materials, and the overall architecture. The continuous dialogue between science, engineering, and design -- a core of Empatica's approach -- resulted in a product that will constitute the gold standard for on-wrist physiological signal measurement, and a great platform for researchers in the space industry and in clinical uses. EmbracePlus is a device that is built to enable discovery, both in space and on Earth. Empatica wants to continue to explore space applications after the launch of EmbracePlus. The company is already planning to complement the product offering with the development of specific biomarkers suitable for the detection of different conditions in space, specifically tiered around mental health applications.
Task Progress:	The Translational Research Institute for Space Health (TRISH)/NASA funded project helped Empatica accelerate mechanical and electrical engineering development for the new Empatica wearable device called EmbracePlus (formerly code-named EmbraceX). The subaward was specifically intended to cover part of the tooling, testing, and technology integration expenses incurred during the product prototyping phase, in order to allow EmbracePlus to reach the Engineering Validation (EV) stage of development. EmbracePlus is intended to be the next-generation wearable device for precision physiological signal monitoring. It is a wrist-worn watch equipped with medical grade sensors, that allows for continuous monitoring of electrodermal activity (EDA), accelerometry, heart rate (HR), peripheral oxygen saturation (SpO2), and peripheral temperature parameters. Empatica is planning to integrate different services and capabilities around the device, in order to make the product suitable for the specificity of different use cases already served by the company, including: remote patient monitoring in healthcare applications, especially neurology; performing clinical trials and data collection for enterprise and academic research; and support platform application for wellness monitoring for third party commercial partners. NASA/TRISH contribution has allowed the team to broaden the horizons of possible use of the product, engineering a solution that can be suitable for astronauts' health monitoring during space missions. The device is expected to be commercialized in Q4 2020 (delayed from Q3 2020 due to COVID-19 impact on product development and supply chain) and will include the following features: accelerometer, cutting-edge photoplethysmographic sensors (PPG), ventral EDA, Bluetooth, a breathable and comfortable strap, large memory storage, a low power e-ink display, water-resistant design, and usability in low/zero gravity settings. Empatica is using state-of-the-art sensors, materials, and construction methods to make EmbracePlus a key product that brings the company closer to the development of a holistic, space-ready research platform for the real-time health monitoring of astronauts. Empatica overall work is focused around building reliable Artificial Intelligence (AI) based physiological biomarkers, and EmbracePlus will serve as the foundation for this platform. Additional improvements in this regard will focus on fatigue, pain, sleep, alertness, stress, mood, activity, and sickness/wellness. During the prototyping stage of product development (covered in part by NASA/TRISH funding), the sensor R&D team evaluated the viability of cutting-edge PPG sensors in the current form factor, with accommodation for secondary noise cancellation. Industrial Design (ID) focused on human factors and ergonomics, as well as working with the Mechanical Engineering team (ME) to determine the optimal form factor, the strap/band component and the choice of a suitable display. The ME team focused on running trials on critical or otherwise risky features, generating a non-form-factor computer-aided design (CAD) package to ensure the integration of all the required design elements. Tolerance analyses have been run on critical feature stacks. The Electrical Engineering team (EE) worked on a Non-Form-Factor Schematic and relative bill of materials (BOM). It also worked with the ME team to ensure relative component integration, and with the Sensor R&D team to ensure alignment with sensor requirements. The global disruption caused by the ongoing (as of June 2020) COVID-19 pandemic has challenged the EmbracePlus product development, affecting the timeline initially presented to TRISH/NASA with a 4 month delay. Initially, COVID-19 disrupted Empatica's capability to source key components in China and South Korea during Q1 2020, due to the closure of most key electronics suppliers in the area around January-March 2020. Further delays were experienced during the slow reopening process, due to risk mitigation measures of individual suppliers and newly implemented regulations. Finally, with subsequent national lockdowns in Europe and stay at home advisories in Massachusetts lasting into June 2020, access of key testing equipment to company employees was difficult. Granted these logistical challenges, specific tasks requiring testing equipment were inevitably delayed. However Empatica never stopped its product development activities: the company immediately shifted all its workforce to remote working and was able to ensure the continuity of the EmbracePlus development. Key findings/Major milestones reached within the project perimeter. Overall mechanical engineering system integration effort for the EmbracePlus product (pod design, sensor integration, electronics integration, sensorized band integration) with a focus on: - Environmental sealing. - Sensors selection and testing (Temperature, PPG, EDA). - Band testing. - Display selection and testing.
Bibliography Type:	Description: (Last Updated: 09/01/2020)
Significant Media Coverage	Cision PRNewswire. "Product unveiled to the press at Cision website." Web release, November 2019., Nov-2019